



INDIVIDUAL LOGBOOK: SOFTWARE ARCHITECTURE (CSE3433)

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

Cloud-Native Learning Management System (LMS)

(Cloud-Native / Layered Architecture)

Nur Alia Farisya Binti Mohd Khairy

S75892

1.0 Team Information

- Project Title: Cloud-Native Learning Management System (LMS)
- Group Number: Group 7 (K2)
- Prepared For: Dr. Mohamad Nor Hassan
- Name: Nur Alia Farisya Binti Mohd Khairy (S75892)
- Assigned Role: Architect and DevOps Lead

1.1 Specific Allocation of Duties

Based on how our team worked on project parts:

- Deliverable 1 (Project Proposal): I selected the Selected Architecture Style and defined what our system should do. Aliya wrote down what the system needs to do at first. We all worked together on describing the problem. What each team member does.
- Deliverable 2 (Architecture Design): I wrote the System Overview and made a diagram showing the system context. Aliya did the Functional Requirements, Non-Functional Requirements, finding out what services or components we need, the Interaction Diagram, and the Deployment Diagram.
- Deliverable 3 (Prototype Progress Report): I checked if our system's main architectural parts are working correctly and if it connects to the cloud properly. Aliya wrote the code. Documented how the prototype is going. Zihan tested if the modules work well together at first.
- Deliverable 4 (Project Report and Deliverables): I wrote the Technology Stack analysis. Aliya wrote about how we implemented the prototype. Zihan did the Architecture Evaluation. We all wrote the Conclusion and Reflection part together.

2.0 Weekly Progress and GenAI Usage Record

Phase 1: Project Proposal (Weeks 1 - 5)

- Focus: Core problem analysis, structural goal mapping, and establishing the foundational Three-Layer architecture style parameters for Deliverable 1.

Week 1 - 2 (Problem Identification and Scope Definition)

- Undertaken Tasks: Evaluating infrastructure bottlenecks, studying server degradation patterns, and detailing team task allocations.
- Contributions:
 - Analyzed how traditional localized university portals face performance issues, high latency, and crashes during high-concurrency peak academic periods.
 - Evaluated the tight structural coupling inherent in legacy monolithic apps, mapping out how UI updates often require changing the core backend.
- Challenges: Limiting the system boundary definitions so that our cloud-native blueprint fits a manageable student assignment scope.
- Action Plans: Clearly map the architectural boundaries between the presentation, business logic, and backend storage subsystems.

GenAI Usage Record (Week 1-2)

- Date (Week): Week 2
- Tool Used: Gemini
- Purpose: Refinement of architectural problem descriptions.
- Prompt Used: *"If a university student portal always crashes or lags like crazy during course registration week or when everyone tries to submit a quiz at the same time, what's usually the main issue from an architectural perspective? Can you list down what's wrong with systems like that?"*
- Output Summary: Provided core points regarding monolithic system weaknesses, traffic bottlenecks on localized servers, and database scaling limitations.
- How It Was Used: Adapted and integrated directly into Section 2.0 Problem Description of our team's project report.

Week 3 - 4 (Architecture Style Selection and System Objectives Formulation)

- Undertaken Tasks: Drafting the architecture style selection and writing the clear system objective criteria for the proposal.
- Contributions:

- Formulated our Selected Architecture Style section, establishing the logical rules for a Three-Layer Architecture (Presentation, Application, and Data Access Layers).
- Drafted the System Objectives, focusing on moving from a tightly coupled structure to a modular platform with clear layer separation.
- Challenges: Making sure my structural objectives matched up with the initial functional requirements that Aliya was compiling.
- Action Plans: Review Aliya's functional requirements definitions to ensure they align cleanly with our layer boundaries.

Week 5 (Deliverable 1 Submission: Project Proposal)

- Undertaken Tasks: Reviewing individual parts, checking layout uniformity, and submitting the 3–4 pages Project Proposal.
- Contributions:
 - Reviewed the complete proposal file to make sure my architecture style section and system objectives connected smoothly with Aliya's initial functional requirements.
 - Documented my own team role and task tracking points for the Deliverable 1 submission text.
- Challenges: Keeping formatting and spacing uniform when joining markdown elements from different team files.
- Action Plans: Submit Deliverable 1 on time and set up the Visual Paradigm canvas for Phase 2 diagramming.

Phase 2: Architecture Design and Verification (Weeks 6 - 7)

- Focus: Compiling structural models, mapping system overviews, drawing required diagrams, and finishing the 6–8 pages Architecture Design document.

Week 6 (System Overview Design and Context Modeling)

- Undertaken Tasks: Writing the high-level system summary and modeling context scopes in Visual Paradigm.

- Contributions:
 - Authored the System Overview section, outlining how our system splits concerns into three distinct tiers using an Aiven.io managed MySQL instance.
 - Designed and rendered the formal System Context Diagram, defining how Learners, Educators, and Administrators cross boundaries to use the LMS.
- Challenges: Ensuring my System Context Diagram boundaries perfectly matched the components and services that Aliya was detailing for her charts.
- Action Plans: Meet with Aliya to verify that all system data labels align across our context, component, interaction, and deployment diagrams.

GenAI Usage Record (Week 6)

- Date (Week): Week 6
- Tool Used: Gemini
- Purpose: Verification of interaction sequences.
- Prompt Used: *"If a student wants to login to a 3-layer system that uses Java Servlets and DAOs, how does the step-by-step flow work? Like from the moment they fill out the UI form until the password gets checked in the database, which functions get called first?"*
- Output Summary: Generated a sequential vertical execution path: `inputCredentials()` → `validateLogin()` → `findUserByUsername()` → execute SELECT query → `verifyPassword()`.
- How It Was Used: Used to check our system's interaction steps before Aliya finalized the high-fidelity Sequence Diagram layout.

Week 7 (Deliverable 2 Submission: Architecture Design)

- Undertaken Tasks: Combining diagram fields, editing text paragraphs, and submitting the official Deliverable 2 document.
- Contributions:
 - Integrated my System Overview and System Context Diagram with Aliya's Functional and Non-Functional requirements text and diagrams.

- Reviewed the completed architecture style explanation text to make sure our three-tier boundaries were clearly defined.
- Challenges: Organizing multi-page visual layout sizes within the document text limits.
- Action Plans: Submit Deliverable 2 cleanly on the portal and prepare workspace environments for implementation tracking.

Phase 3: Project Observation 1 and Prototype Progress (Weeks 8 - 11)

- Focus: Configuring environment variables, auditing initial code structures, and tracking baseline component connectivity.

Week 8 (Technology Stack Finalization and Infrastructure Setup)

- Undertaken Tasks: Outlining our system layer frameworks and preparing local server configurations.
- Contributions:
 - Wrote our definitive Technology Stack matrix, providing the architectural justification for choosing HTML5, CSS3, JS, Java Servlets, JSP, and MySQL hosted on Aiven.io.
 - Verified the baseline JDBC driver settings to ensure our servlet code could talk securely with the remote cloud nodes.
- Challenges: Resolving port conflicts on our local test setups when booting up Tomcat web instances.
- Action Plans: Modify the default server connection configurations to clear up our development environment.

GenAI Usage Record (Week 8)

- Date (Week): Week 8
- Tool Used: Gemini
- Purpose: Technical technology stack justification generation.

- Prompt Used: *"Is there any actual design benefit to using client-side JavaScript for validation on the frontend if we're already using a managed cloud database like Aiven.io on the backend? Just wondering how it helps with server bandwidth or compute power."*
- Output Summary: Explained that client-side validation screens out empty or malformed requests early, saving the application layer from wasting compute resources processing dead requests before hitting the database cluster.
- How It Was Used: Incorporated as our main engineering rationale within Section 6.0 Technology Stack of the report.

Week 9 (Project Observation 1 - Milestone Evaluation)

- Undertaken Tasks: Demonstrating early interface controls and verifying vertical code communication loops.
- Contributions:
 - Led the live milestone presentation, showing how form inputs are handled across our different architectural layers without bypassing security controls.
 - Defended our logical layer structure before the examiner, showing how the DAO pattern handles raw queries cleanly.
- Challenges: Explaining our structural approach clearly within the presentation time limits.
- Action Plans: Map out clear source code structures to simplify our future prototype reviews.

Week 10 (Backend Core Architecture Auditing)

- Undertaken Tasks: Reviewing request parameters, checking controller files, and auditing safety controls.
- Contributions:
 - Audited the core application code patterns (AuthServlet.java and UserDAO.java) to ensure query structures were built securely using parameterized styles.

- Checked the DatabaseConnection.java file to ensure connection string endpoints linked safely to the remote cloud engine.
- Challenges: Fixing raw exception blocks that exposed sensitive internal variables directly on the interface screens.
- Action Plans: Help clean up the backend error logs so they output user-friendly messages instead of raw system variables.

GenAI Usage Record (Week 10)

- Date (Week): Week 10
- Tool Used: Gemini
- Purpose: Code review and security assessment.
- Prompt Used: *"If my teammate wrote a query using a PreparedStatement with .setString(), does that completely block SQL injection attacks? How does that mechanism actually work under the hood to keep it secure?"*
- Output Summary: Confirmed that PreparedStatement parameterizes user inputs, treating them strictly as literal strings rather than executable SQL code, which neutralizes injection risks.
- How It Was Used: Used this security analysis to support our data layer defense sections.

Week 11 (Deliverable 3 Submission: Prototype Progress Report)

- Undertaken Tasks: Verifying our live GitHub repositories and checking cloud database tables.
- Contributions:
 - Checked the live data metrics on our Aiven.io Cloud Database Live Data Viewer screen to verify that registration records were saving correctly.
 - Supported Aliya with her prototype implementation section compilation by organizing screenshots of our working modules.
- Challenges: Keeping our active software code updates in sync with our static system charts.

- Action Plans: Freeze our current master software branch and launch our final evaluation phase.

Phase 4: Observation 2 and Final Report Compilation (Weeks 12 - 13)

- Focus: Tracking evaluation parameters, writing joint conclusion blocks, and finalizing our 24-page report alongside our 10–12 pages Individual Logbooks.

Week 12 (Joint Conclusion Preparation and Assessment Reviews)

- Undertaken Tasks: Reviewing quality attributes and drafting our combined reflection blocks.
- Contributions:
 - Worked with Zihan to review her architecture evaluation table, checking the maintainability, scalability, and availability ratings.
 - Collaborated with Aliya and Zihan to draft our shared Section 10.0 Conclusion and Reflection blocks, documenting how our three-layer setup successfully avoids common anti-patterns.
- Challenges: Writing a cohesive conclusion that balanced our different perspectives on frontend, backend, and cloud design.
- Action Plans: Organize a final editing sweep to align our writing style before the submission deadline.

GenAI Usage Record (Week 12)

- Date (Week): Week 12
- Tool Used: Gemini
- Purpose: Explaining physical vs. logical co-location.
- Prompt Used: *"How do I explain to my lecturer if our controller and business logic code sit in the exact same physical Servlet file, but we are still treating them as separate logical layers? I don't want the lecturer to think our system architecture is messy."*

- Output Summary: Introduced the concept of physical co-location versus logical separation, explaining that keeping them together reduces deployment latency while maintaining clean modular duties.
- How It Was Used: Placed directly into Section 4.0 Architecture Design to explain our unified tier processing logic.

Week 13 (Deliverable 4 Submission: Final Report and Logbook)

- Undertaken Tasks: Assembling all text chapters, formatting code blocks, and submitting Deliverable 4.
- Contributions:
 - Compiled the final 24-page report, verifying that my technology stack write-up and our joint conclusion section integrated smoothly with Aliya's prototype chapters and Zihan's evaluation matrix.
 - Finished this individual logbook, outlining my duties and transparently listing my AI records to meet our academic integrity guidelines.
- Challenges: Managing large text files, image sizes, and code listings right before our June 19th, 2026, deadline.
- Action Plans: Upload the final report and individual logbooks to the university submission site on time.

GenAI Usage Record (Week 13)

- Date (Week): Week 13
- Tool Used: Gemini
- Purpose: Quality Attribute Analysis refinement.
- Prompt Used: *"Help our team compile a comprehensive quality attribute table evaluation for our LMS report. We need to evaluate Maintainability, Scalability, Availability, Security, Portability, and Modularity based on a logical 4-layer structure."*
- Output Summary: Generated a dense evaluation matrix classifying key architectural properties alongside technical justifications.

- How It Was Used: Used to review and cross-check the properties listed in Section 8.0 Architecture Evaluation Table.

Phase 5: Final Demonstration (Week 14)

- Focus: Preparing slide structures, managing presentation practice runs, and completing the final evaluation panel presentation.

Week 14 (Deliverable 5: Final Prototype Showcase)

- Undertaken Tasks: Rehearsing project slides, testing our live system connections, and completing our 12–15 minute Deliverable 5 presentation.
- Contributions:
 - Delivered the opening slides of our group pitch, presenting our high-level System Overview and detailing our architecture choices to the evaluation panel.
 - Answered the panel's technical questions regarding how our multi-tier layout and Aiven.io database engine help solve legacy system constraints.
- Challenges: Explaining complex architectural trade-offs within our strict presentation time limit.
- Action Plans: Conclude the Software Architecture module successfully, having demonstrated a stable, maintainable cloud-native blueprint.

GenAI Usage Record (Week 14)

- Date (Week): Week 14
- Tool Used: Gemini
- Purpose: Presentation speech formulation.
- Prompt Used: *"I have to present our system overview and architecture style next week in front of a panel. Any tips on how to structure the intro and explain the layer division clearly so it sounds smooth and confident."*
- Output Summary: Suggested a high-level delivery structure focusing heavily on resolving legacy portal flaws.

- How It Was Used: Used as my personal speaking guide during our mock presentation practice runs.